Name: _____

WORKSHEET

Straight-line motion 2

1 Find the antiderivatives of the following functions:

a $f(x) = 3x^2 - x + 5$

b $f(t) = 12t - 5t^2$

c $\frac{dy}{dx} = x^2 + 6x - 3$

d $v'(t) = 10t - 5$

2 The velocity of an object is given by $v(t) = -4t + 3$ m/s. Find:

a the displacement after 2 seconds if $s(0) = 5$.

b the initial displacement if $s(8) = 20$.

c the displacement after 5 seconds if $s(3) = 1$.

d when the object is at $s(8) = -10$ m, if it is initially at the origin.

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3 An object accelerates at $a(t) = 5 \text{ m/s}^2$ for all $t \geq 0$. Find:

a the velocity after 3 seconds if it starts from rest.

b the velocity after 7 seconds if $v(0) = 10$ m/s.

c when the object moves with $v(t) = 25$ m/s, if $v(0) = -5$ m/s.

d when the object moves with $v(t) = 12$ m/s, if $v(5) = 3$ m/s.

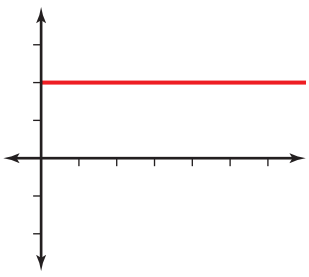
e the displacement after 2 seconds if the object is initially at rest at the origin.

f the displacement after 5 seconds if the object is moving at $v(1) = 10$ m/s and is initially at $s(0) = -5$ m.

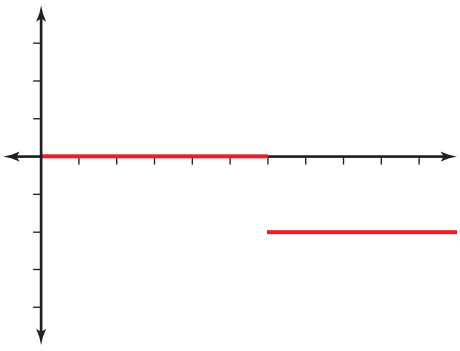
g when the object is at $s(t) = 200$ m if the object is moving with $v(0) = 15$ m/s at the origin

4 Given the following velocity time graphs, sketch a displacement time graph.

a

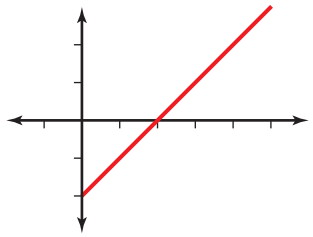


b

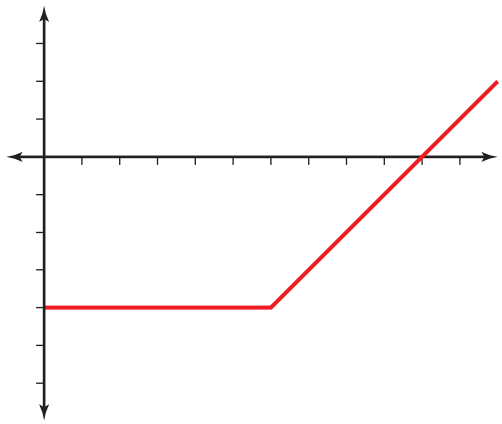


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c

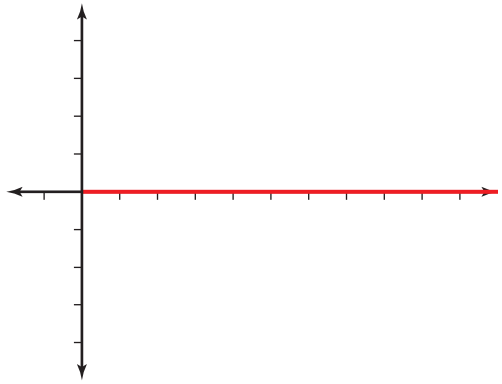


d

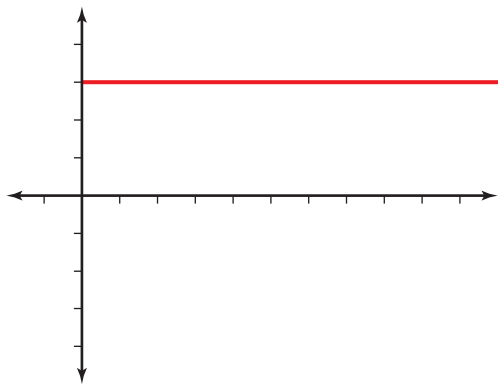


5 Given the following acceleration time graphs, sketch a velocity time graph and a displacement time graph.

a

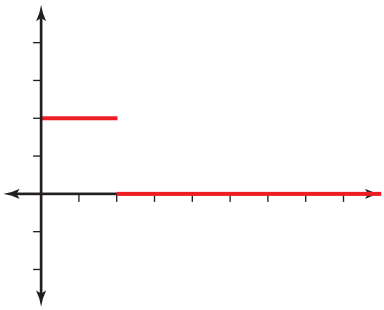


b



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c



Answers

1 **a** $F(x) = x^3 - \frac{x^2}{2} + 5x + c$

b $F(t) = 6t^2 - \frac{5t^3}{3} + c$

c $y = \frac{x^3}{3} + 3x^2 - 3x + c$

d $v(t) = 5t^2 - 5t + c$

2 **a** $s(2) = 3m$

b $s(0) = 124m$

c $s(5) = -25m$

d $t = 3.1s$

3 **a** $v(3) = 15$ m/s

b $v(7) = 45$ m/s

c $t = 6s$

d $t = 6.8s$

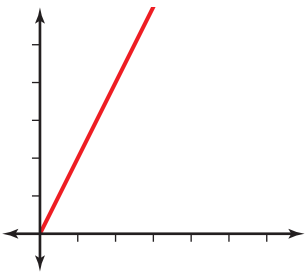
e $s(2) = 10m$

f $s(5) = 82.5m$

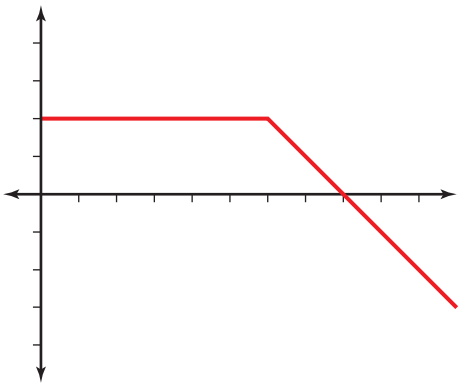
g $t \approx 6.4s$

4 The following are examples of correct solutions (Changes to slope and vertical shift may be acceptable).

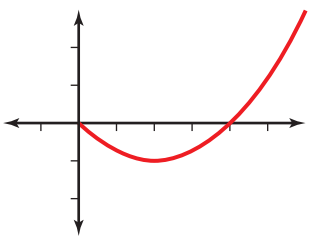
a



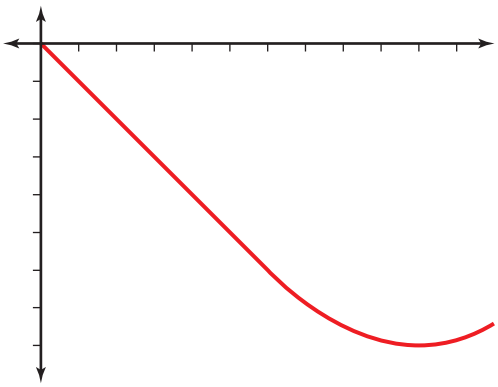
b



c

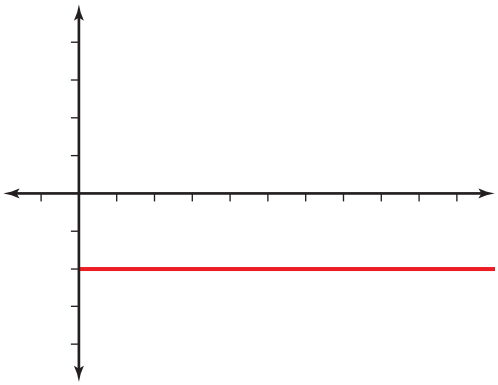


d

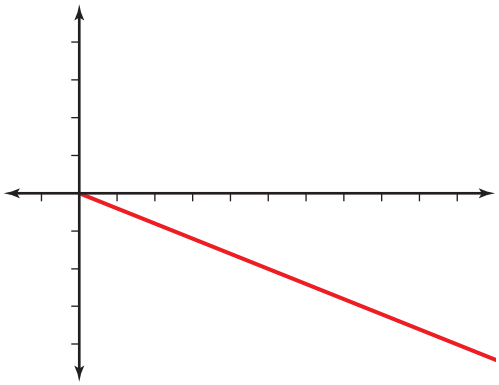


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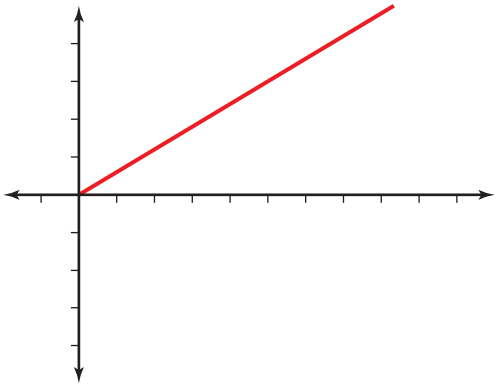
5 **a** vel



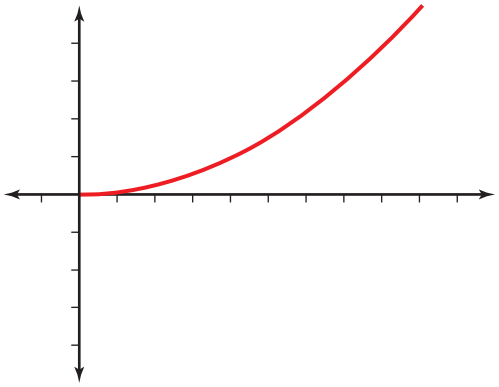
disp



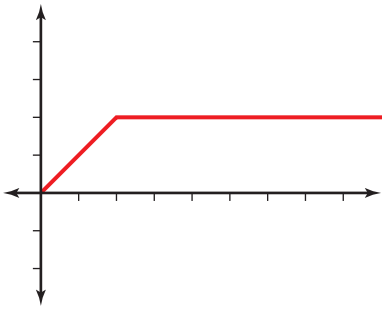
b vel



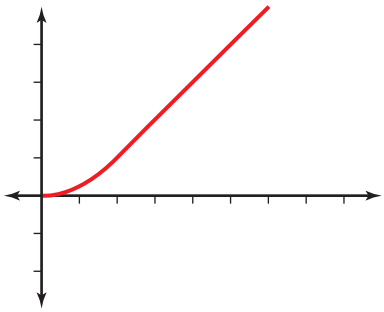
disp



c vel



disp



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